Practical No. 12: *Implement program to use machine-learning model on given dataset.

X. Conclusion

The Decision Tree model was used on the Iris dataset and it worked very well. The model correctly predicted all the test data. Decision trees are easy to understand because they show the steps of decision-making in a tree structure. They can be used for both classification (categories) and regression (numbers). This practical shows how decision trees can help make decisions from data clearly and accurately.

XI. Practical Questions

1. **What is a Decision Tree?**

   A Decision Tree is a model that splits data step by step based on features, forming a tree. Each branch is a decision, and the leaves show the final result.
2. **Two real-world uses of decision trees:**
   - Predicting diseases in medical diagnosis
   - Predicting if a customer will leave a company (churn prediction)
3. **What is Gini Index?**

   It measures how mixed or pure the data is in a group. Lower Gini means the data in that group is mostly the same class.
4. **What is Entropy?**

   Entropy measures uncertainty or disorder in a group. If all data belongs to the same class, entropy is 0.
5. **What is Information Gain?**

   It tells us how much a split reduces uncertainty. The split with the highest information gain is chosen for the tree.
6. **Why do we prune decision trees?**

   To remove extra branches and make the tree simpler, so it works better on new data.
7. **Difference between Gini Index and Entropy:**

Feature	Gini Index	Entropy
Tells us	How mixed/pure the group is	How uncertain/messy it is
Value range	0 = pure, 0.5 = mixed	0 = pure, 1 = messy
Goal	Minimize Gini	Minimize Entropy
Example	10 Flu + 0 Cold → 0	5 Flu + 5 Cold → 1
Used in	CART algorithm (fast)	ID3 / C4.5 algorithms

XII. Exercises

1. Modify the above program to use criterion="entropy" instead of "gini".

```
dt_classifier = DecisionTreeClassifier(criterion="entropy",
random_state=42)
```

2. Build a simple decision-support system for medical diagnosis using a decision tree.

```python
import pandas as pd
from sklearn.tree import DecisionTreeClassifier
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score

# ========================
# Medical dataset
# 1 = Yes, 0 = No for symptoms
data_med = {
    'Fever':   [1, 1, 0, 1, 0, 0, 1, 0, 1, 0, 1, 0],
    'Cough':   [1, 0, 1, 1, 0, 1, 1, 0, 1, 1, 0, 0],
    'Fatigue': [1, 1, 0, 1, 0, 0, 0, 1, 1, 0, 1, 0],
    'Diagnosis': [
        'Flu', 'Flu', 'Cold', 'Flu', 'Healthy', 'Cold',
        'Flu', 'Cold', 'Flu', 'Healthy', 'Flu', 'Cold'
    ]
}


# Create DataFrame
df_med = pd.DataFrame(data_med)
print(df_med)
# Features (inputs) and target (output)
X = df_med[['Fever', 'Cough', 'Fatigue']]
y = df_med['Diagnosis']

# Split data into training and testing sets (70% train, 30% test)
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)

# Initialize Decision Tree Classifier with Entropy
dt = DecisionTreeClassifier(criterion='entropy', random_state=42)

# Train the model
dt.fit(X_train, y_train)

# Predict on test data
y_pred = dt.predict(X_test)

# Evaluate the model
accuracy = accuracy_score(y_test, y_pred)

# Display results
print("=== Medical Diagnosis System ===")
print(f"Accuracy: {accuracy:.2f}")
print("Predicted Diagnoses:", y_pred)
```

Output

```
    Fever  Cough  Fatigue Diagnosis
0       1      1        1       Flu
1       1      0        1       Flu
2       0      1        0      Cold
3       1      1        1       Flu
4       0      0        0   Healthy
5       0      1        0      Cold
6       1      1        0       Flu
7       0      0        1      Cold
8       1      1        1       Flu
9       0      1        0   Healthy
10      1      0        1       Flu
11      0      0        0      Cold
=== Medical Diagnosis System ===
Accuracy: 0.75
Predicted Diagnoses: ['Flu' 'Cold' 'Flu' 'Flu']
```

Insight

- **Dataset:** 12 patients with symptoms Fever, Cough, Fatigue.
- **Model Accuracy:** 75%
- **Predictions on test data:** Flu, Cold, Flu, Flu
- **Decision pattern:**
    - Fever = 1 → usually Flu
    - Fever = 0 & Cough = 1 → usually Cold
    - Fever = 0 & Cough = 0 → Healthy

The Decision Tree can classify patients based on symptoms with good accuracy.